In section 33.2.1 I described `HttpClient` as consisting of a pipeline of handlers. The big advantage of this pipeline, much like the middleware pipeline of your application, is that it allows you to add cross-cutting concerns to all requests. For example, `IHttpClientFactory` automatically adds a handler to each `HttpClient` that logs the status code and duration of each outgoing request.

In addition to logging, another common requirement is to handle transient errors when calling an external API. Transient errors can happen when the network drops out, or if a remote API goes offline temporarily. For transient errors, simply trying the request again can often succeed, but having to write the code to do so manually is cumbersome.

ASP.NET Core includes a library called Microsoft.Extensions.Http.Polly that makes handling transient errors easier. It uses the popular open-source library Polly (https://github.com/App-vNext/Polly) to automatically retry requests that fail due to transient network errors.

Polly is a mature library for handling transient errors that includes a variety of error-handling strategies, such as simple retries, exponential backoff, circuit breaking, and bulkhead isolation. Each strategy is explained in detail at https://github.com/App-vNext/Polly, so be sure to read about the benefits and trade-offs when selecting a strategy.

To provide a taste of what's available, we'll add a simple retry policy to the `ExchangeRatesClient` shown in section 33.2. If a request fails due to a network problem, such as a timeout or a server error, we'll configure Polly to automatically retry the request as part of the handler pipeline, as shown in figure 33.5.

